

Data Analytics Capstone

**Machine Learning-Based Prediction of Urban Air Quality Using Environmental and  
Meteorological Data**

**Capstone Project Synopsis**

Vinita R. Shetty

Walsh College

**QM640: Data Analytics Capstone**

Mentor: Karthik Umesh

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## **Introduction**

### **Background and Context**

Air pollution has emerged as one of the most critical environmental and public health challenges worldwide. Rapid urbanization, industrial expansion, population growth, and increased vehicular emissions have significantly contributed to deteriorating air quality, particularly in densely populated urban regions. According to the World Health Organization (WHO, 2021), exposure to air pollution is responsible for millions of premature deaths annually, making effective air quality management a global priority. Governments and environmental agencies are increasingly investing in continuous monitoring systems and data-driven approaches to better understand pollution patterns and support evidence-based policy decisions.

Among the various air pollutants, fine particulate matter (PM<sub>2.5</sub>) is considered one of the most hazardous due to its ability to penetrate deep into the human respiratory system and bloodstream. Prolonged exposure to elevated PM<sub>2.5</sub> concentrations has been associated with respiratory diseases, cardiovascular disorders, reduced lung function, and increased mortality. Since PM<sub>2.5</sub> concentrations are influenced by multiple environmental and meteorological factors, understanding these relationships is essential for improving air quality forecasting and implementing effective pollution control measures. Previous studies conducted in Beijing have also demonstrated the significant influence of meteorological conditions on PM<sub>2.5</sub> concentrations (Zhang et al., 2017).

Traditional air quality monitoring systems primarily provide real-time observations but have limited capability to accurately predict future pollution levels. Recent advances in machine learning have enabled researchers to develop predictive models capable of learning complex relationships between meteorological variables, pollutant concentrations, and temporal factors. These models can improve forecasting accuracy, support early warning systems, and assist policymakers, environmental agencies, healthcare organizations, and urban planners in making informed decisions (Hastie et al., 2021; Zhou, 2021).

This study proposes the use of machine learning techniques to predict PM<sub>2.5</sub> concentrations using environmental and meteorological variables. Multiple machine learning regression models, including Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost, will be developed and compared to identify the most accurate predictive model (Bishop, 2006; Breiman, 2001; Friedman, 2001; Chen & Guestrin, 2016). The findings are expected to improve air quality forecasting and support evidence-based environmental decision-making for sustainable urban environmental management.

### **Problem Statement**

Air pollution has become a significant environmental and public health concern, particularly in rapidly urbanizing regions where pollutant concentrations frequently exceed recommended safety limits. Among the various air pollutants, fine particulate matter (PM<sub>2.5</sub>) poses one of the greatest health risks because prolonged exposure is associated with respiratory illnesses, cardiovascular diseases, and premature mortality. Although existing air quality monitoring systems provide continuous measurements of pollutant concentrations, they primarily

report current conditions and have limited capability to accurately predict future PM<sub>2.5</sub> levels or explain the complex relationships between environmental and meteorological factors that influence air quality. This limitation restricts the ability of environmental agencies, policymakers, and urban planners to implement timely interventions and effective pollution control strategies. Therefore, there is a need to develop reliable machine learning-based predictive models that can improve PM<sub>2.5</sub> forecasting and support evidence-based environmental decision-making (Breiman, 2001; Chen & Guestrin, 2016).

### **Purpose of the Study**

The purpose of this study is to develop and evaluate machine learning models for predicting PM<sub>2.5</sub> concentration using environmental and meteorological variables obtained from the Beijing Multi-Site Air Quality Dataset available through the UCI Machine Learning Repository. Multiple regression-based machine learning algorithms, including Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, and XGBoost Regression, will be developed and compared to identify the model with the best predictive performance. Model performance will be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ( $R^2$ ). The findings are expected to support environmental agencies, policymakers, and urban planners in improving air quality forecasting and developing data-driven strategies for sustainable environmental management (Pedregosa et al., 2011).

## Scope and Objectives

### Scope of the Study

The scope of this study is to develop and evaluate machine learning models for predicting PM<sub>2.5</sub> concentration using environmental and meteorological variables obtained from the Beijing Multi-Site Air Quality Dataset available through the UCI Machine Learning Repository. The study focuses on analyzing 420,768 historical hourly observations collected from 12 nationally controlled air quality monitoring stations in Beijing, China, between 1 March 2013 and 28 February 2017(Chen, 2019) to identify patterns and relationships between air pollutants, meteorological conditions, and PM<sub>2.5</sub> concentration.

The study includes data preprocessing, exploratory data analysis, feature selection, model development, model evaluation, and interpretation of results using the selected secondary dataset. The study does not involve real-time data collection or deployment of a live prediction system. As the dataset is publicly available and contains no personally identifiable information, the study presents minimal ethical concerns. Since the models are developed and validated using data collected only from Beijing, the findings may require further validation before being generalized to other geographical regions. The proposed work will be completed within the ten-week capstone period using Python and standard machine learning libraries (Pedregosa et al., 2011). A proposed 10-week project timeline is provided in Appendix D to demonstrate the feasibility of completing the study within the prescribed capstone duration.

## Research Objectives

The primary objective of this study is to develop an accurate and reliable machine learning model for predicting PM2.5 concentration using environmental and meteorological variables. The specific objectives are:

1. To identify the environmental and meteorological variables that significantly influence PM2.5 concentration.
2. To analyze the relationship between environmental variables and PM2.5 concentration using exploratory data analysis and statistical techniques.
3. To develop and compare multiple machine learning regression models for predicting PM2.5 concentration.
4. To evaluate the predictive performance of the developed models using appropriate regression evaluation metrics.
5. To recommend the most suitable machine learning model for supporting air quality forecasting and environmental decision-making.

## Research Questions

The study seeks to answer the following research questions:

- **RQ1.** Which environmental and meteorological variables significantly influence PM2.5 concentration?
- **RQ2.** How accurately can machine learning models predict PM2.5 concentration using environmental and meteorological variables?

- **RQ3.** Which machine learning regression model provides the best predictive performance for PM2.5 concentration?
- **RQ4.** Which environmental and meteorological variables contribute the most to the predictive performance of the best-performing machine learning model?

The study is supported by null and alternative hypotheses corresponding to each research question. A detailed mapping of the research questions and their associated hypotheses is presented in Appendix C (Table C1).

### Sample Size Calculation

The minimum sample size required for this study was determined using the Confidence Interval (CI) method. Since the study utilizes a large publicly available secondary dataset and all four research questions are addressed using the same dataset, the Confidence Interval method was considered appropriate for determining the minimum sample size.

The sample size was calculated using the following formula:

$$n = \frac{Z^2 \times p(1 - p)}{e^2}$$

Where,  $n$  = minimum required sample size,  $Z$  = Z-score (1.96 for 95% confidence),  $p$  = estimated population proportion (0.50), and  $e$  = margin of error (0.05).

Using these parameters:

$$n = \frac{(1.96)^2 \times 0.50 \times (1 - 0.50)}{(0.05)^2}$$

$$n = \frac{3.8416 \times 0.25}{0.0025}$$

$$n = 384.16 \approx 385$$

Therefore, the minimum required sample size is 385 observations.

Since all four research questions utilize the same secondary dataset and involve regression-based machine learning analyses, the same Confidence Interval assumptions (95% confidence level, 5% margin of error, and estimated population proportion of 0.50) were applied to determine the minimum sample size for each research question. Therefore, the calculated minimum sample size is identical for RQ1 to RQ4.

Table 1 presents the minimum sample size calculation for each research question using the Confidence Interval method.

**Table 1**

*Minimum Sample Size Calculation for Research Questions*

| Research Question | Method Used         | Key Parameters                 | Minimum Sample Size (N) |
|-------------------|---------------------|--------------------------------|-------------------------|
| RQ1               | Confidence Interval | $Z = 1.96, p = 0.50, e = 0.05$ | 385                     |
| RQ2               | Confidence Interval | $Z = 1.96, p = 0.50, e = 0.05$ | 385                     |
| RQ3               | Confidence Interval | $Z = 1.96, p = 0.50, e = 0.05$ | 385                     |
| RQ4               | Confidence Interval | $Z = 1.96, p = 0.50, e = 0.05$ | 385                     |

### Final Sample Size Decision

The Confidence Interval method indicates a minimum sample size requirement of 385 observations for each research question. Since all four research questions will be addressed using the same secondary dataset, the final minimum sample size remains 385 observations. However,



the Beijing Multi-Site Air Quality Dataset contains 420,768 hourly observations, which substantially exceeds the minimum sample size requirement. Therefore, the entire dataset will be used for data preprocessing, model development, validation, and evaluation. Utilizing the complete dataset improves the robustness of the machine learning models and enhances the reliability of the study findings.

### Data Dictionary

Table 2 summarizes the key variables used in this study. The variables include the dependent variable (PM<sub>2.5</sub>) and the environmental and meteorological variables that will be used as predictors during model development and evaluation.

**Table 2**

*Data Dictionary for the Beijing Multi-Site Air Quality Dataset*

| Variable                | Description                                                                             | Data Type | Role                               |
|-------------------------|-----------------------------------------------------------------------------------------|-----------|------------------------------------|
| <b>PM<sub>2.5</sub></b> | Fine particulate matter concentration ( $\mu\text{g}/\text{m}^3$ )                      | Numeric   | <b>Target (Dependent Variable)</b> |
| PM <sub>10</sub>        | Particulate matter ( $\leq 10 \mu\text{m}$ ) concentration ( $\mu\text{g}/\text{m}^3$ ) | Numeric   | Environmental Predictor            |
| SO <sub>2</sub>         | Sulfur dioxide concentration ( $\mu\text{g}/\text{m}^3$ )                               | Numeric   | Environmental Predictor            |
| NO <sub>2</sub>         | Nitrogen dioxide concentration ( $\mu\text{g}/\text{m}^3$ )                             | Numeric   | Environmental Predictor            |
| CO                      | Carbon monoxide concentration ( $\text{mg}/\text{m}^3$ )                                | Numeric   | Environmental Predictor            |
| O <sub>3</sub>          | Ozone concentration ( $\mu\text{g}/\text{m}^3$ )                                        | Numeric   | Environmental Predictor            |
| TEMP                    | Air Temperature ( $^{\circ}\text{C}$ )                                                  | Numeric   | Meteorological Predictor           |
| PRES                    | Atmospheric pressure (hPa)                                                              | Numeric   | Meteorological Predictor           |

| Variable | Description                                  | Data Type   | Role                     |
|----------|----------------------------------------------|-------------|--------------------------|
| DEWP     | Dew point temperature (°C)                   | Numeric     | Meteorological Predictor |
| RAIN     | Rainfall (mm)                                | Numeric     | Meteorological Predictor |
| WSPM     | Wind speed (m/s)                             | Numeric     | Meteorological Predictor |
| wd       | Wind direction (Categorical: N, NE, E, etc.) | Categorical | Meteorological Predictor |

Table 2 presents the principal variables used in this study. A comprehensive data dictionary containing all dataset variables, units of measurement, and additional metadata will be included in the Interim and Final Reports to support complete documentation and reproducibility.

### Data Description

#### Data Source

The study utilizes the Beijing Multi-Site Air Quality Dataset (Chen, 2019) obtained from the UCI Machine Learning Repository, a widely recognized repository of machine learning datasets used for academic and research purposes. The dataset contains historical hourly air quality and meteorological observations collected from 12 nationally controlled air quality monitoring stations in Beijing, China, covering the period from 1 March 2013 to 28 February 2017 (Chen, 2019).

The dataset consists of 420,768 hourly observations across 18 variables, including air pollutant concentrations, meteorological measurements, temporal variables, and monitoring

station identifiers. The Beijing Multi-Site Air Quality Dataset was selected because it is a publicly available, benchmark dataset that provides integrated hourly air quality and meteorological observations in a single, well-documented resource. The dataset satisfies the data requirements of this study by including the target variable (PM<sub>2.5</sub>), relevant predictor variables, and sufficient historical observations to develop, compare, and evaluate machine learning regression models. Furthermore, its widespread use in academic research supports reproducibility and facilitates comparison with existing studies.

### **Dataset Description**

Each observation in the dataset represents the air quality and meteorological conditions recorded at a specific monitoring station during a particular hour. The dataset contains temporal variables (year, month, day, and hour), air pollutant measurements (PM<sub>2.5</sub>, PM<sub>10</sub>, SO<sub>2</sub>, NO<sub>2</sub>, CO, and O<sub>3</sub>), meteorological variables (temperature, atmospheric pressure, dew point, rainfall, wind direction, and wind speed), and the corresponding monitoring station identifier.

Preliminary exploration of the dataset confirmed that it contains 420,768 hourly observations collected from 12 monitoring stations across Beijing. The dataset also contains missing values in several pollutant and meteorological variables, which will be addressed during the data preprocessing stage before model development. The comprehensive combination of pollution and weather variables makes the dataset suitable for developing regression-based machine learning models for predicting PM<sub>2.5</sub> concentrations (Chen, 2019).

## **Dataset Source**

- UCI Machine Learning Repository:

<https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data>

- GitHub Repository:

<https://github.com/ShettyVini15/QM640-Air-Quality-Capstone>

The repository is publicly accessible and contains the project README, dataset source information, repository structure, and supporting files required to reproduce the study. A screenshot of the repository structure is provided in Appendix B (Figure B1).

## **Dataset Preview**

Selected dataset outputs are provided in Appendix A to demonstrate dataset accessibility, structure, dimensions, descriptive statistics, and data quality.

## **Data Availability and Ethical Considerations**

The Beijing Multi-Site Air Quality Dataset is publicly available through the UCI Machine Learning Repository and is intended for research and educational purposes. The dataset contains only environmental monitoring information and does not include personally identifiable or confidential information. Therefore, the study presents minimal ethical concerns related to privacy and confidentiality.

The dataset will be used solely for academic research, and appropriate acknowledgment and citation of the original data source will be provided in accordance with APA 7 guidelines. The use of a publicly available benchmark dataset also supports transparency, reproducibility, and comparability of the research findings.

### Analytic Approach

**Table 3** summarizes the proposed analytical methods, justification, and expected outputs for each research question.

**Table 3**

*Mapping of Research Questions to Analytical Methods*

| Research Question                                                                                                                                             | Proposed Analytical Method                                                         | Justification                                                                                                                               | Expected Output                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>RQ1.</b> Which environmental and meteorological variables significantly influence PM2.5 concentrations?                                                    | Exploratory Data Analysis (EDA), Correlation Analysis, Feature Importance Analysis | To identify relationships between predictor variables and PM2.5 concentration and determine the variables that have the greatest influence. | Identification of the most significant environmental and meteorological predictors of PM2.5 concentration. |
| <b>RQ2.</b> How accurately can machine learning models predict PM2.5 concentration using environmental and meteorological variables?                          | Multiple Machine Learning Regression Models                                        | To evaluate the predictive capability of machine learning algorithms for estimating PM2.5 concentration.                                    | Predicted PM2.5 values and model performance metrics.                                                      |
| <b>RQ3.</b> Which machine learning regression model provides the best predictive performance for PM2.5 concentration?                                         | Comparative Performance Evaluation using Regression Metrics                        | To compare the performance of different regression algorithms and identify the best-performing model.                                       | Selection of the model with the highest predictive performance based on evaluation metrics.                |
| <b>RQ4.</b> Which environmental and meteorological variables contribute the most to the predictive performance of the best-performing machine learning model? | Feature Importance Analysis and SHAP (SHapley Additive exPlanations)               | To determine the relative contribution of predictor variables in the selected machine learning model.                                       | Ranking of predictor variables based on their contribution to model performance.                           |

## **Research Design**

This study adopts a quantitative, predictive, and experimental research design using a publicly available secondary dataset. A supervised machine learning regression approach will be employed to predict PM2.5 concentration. The outcome (target) variable is PM2.5 concentration, while the predictor variables include environmental, meteorological, and temporal variables obtained from the Beijing Multi-Site Air Quality Dataset. (Hastie et al., 2021).

## **Data Preprocessing**

The dataset will be preprocessed by handling missing values, duplicate records, and potential outliers and appropriate feature encoding where required. Exploratory Data Analysis (EDA) will then be performed to assess data quality and understand variable relationships before model development.

## **Model Validation**

The dataset will be divided into training (80%) and testing (20%) subsets to evaluate predictive performance on unseen observations. In addition, 5-fold cross-validation will be employed during model development to improve model reliability and reduce the risk of overfitting (Hastie et al., 2021).

## **Proposed Solution**

The proposed study will employ a supervised machine learning regression approach to predict PM2.5 concentration using environmental and meteorological variables obtained from the Beijing Multi-Site Air Quality Dataset. The analytical workflow will begin with data acquisition and preprocessing, including handling missing values, identifying duplicate records,

and examining outliers where appropriate. Exploratory Data Analysis (EDA) will be conducted to understand the distribution of variables, identify trends and relationships, and assess correlations between predictor variables and PM2.5 concentration.

Relevant features will be selected based on statistical analysis and domain knowledge. The dataset will be divided into 80% training and 20% testing subsets, and model performance will be further validated using 5-fold cross-validation to improve model reliability and reduce the risk of overfitting. Multiple machine learning regression algorithms, including Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, and XGBoost Regression, will be developed and compared to determine the most suitable model for predicting PM2.5 concentration.

Model performance will be evaluated using MAE, RMSE, and  $R^2$ . The best-performing model will be selected based on these metrics. Feature importance analysis will identify the most influential predictors of PM2.5, and SHAP visualizations will be used, where appropriate, to improve model interpretability. The findings are expected to support evidence-based environmental planning and air quality management (Pedregosa et al., 2011).

### Evaluation Metrics

The evaluation metrics used in this study are defined as follows:

- **Mean Absolute Error (MAE):** measures the average absolute difference between the actual and predicted PM2.5 values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Root Mean Squared Error (RMSE)** measures the square root of the average squared differences between the actual and predicted PM2.5 values, giving greater weight to larger prediction errors.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Coefficient of Determination ( $R^2$ )** measures the proportion of variance in PM2.5 concentration explained by the prediction model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where,  $n$  = number of observations,  $y_i$  = actual PM2.5 value,  $\hat{y}_i$  = predicted PM2.5 value, and  $\bar{y}$  = mean of the actual PM2.5 values.

### Recommendation and Applications

The proposed machine learning framework is expected to support environmental protection agencies, policymakers, urban planners, public health organizations, and researchers by improving PM2.5 forecasting and enabling data-driven environmental decision-making. Accurate predictions can support early warning systems, timely public health advisories, pollution mitigation strategies, and sustainable urban planning. The framework can also assist policymakers in developing evidence-based environmental policies and help researchers better understand the influence of environmental and meteorological factors on air quality. Although the framework is developed using data from Beijing, it can be validated and adapted for application in other cities and geographical regions (Pedregosa et al., 2011; Zhou, 2021).



## References

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## Appendix A

### Dataset Verification and Preliminary Exploration

The following figures present selected outputs from the Python notebook to verify dataset accessibility, structure, dimensions, descriptive statistics, and data quality prior to model development.

[6] ✓ 0s df.head()

|   | No | year | month | day | hour | PM2.5 | PM10 | SO2 | NO2 | CO    | O3   | TEMP | PRES   | DEWP  | RAIN | wd  | WSPM | station  |
|---|----|------|-------|-----|------|-------|------|-----|-----|-------|------|------|--------|-------|------|-----|------|----------|
| 0 | 1  | 2013 | 3     | 1   | 0    | 4.0   | 4.0  | 3.0 | NaN | 200.0 | 82.0 | -2.3 | 1020.8 | -19.7 | 0.0  | E   | 0.5  | Dingling |
| 1 | 2  | 2013 | 3     | 1   | 1    | 7.0   | 7.0  | 3.0 | NaN | 200.0 | 80.0 | -2.5 | 1021.3 | -19.0 | 0.0  | ENE | 0.7  | Dingling |
| 2 | 3  | 2013 | 3     | 1   | 2    | 5.0   | 5.0  | 3.0 | 2.0 | 200.0 | 79.0 | -3.0 | 1021.3 | -19.9 | 0.0  | ENE | 0.2  | Dingling |
| 3 | 4  | 2013 | 3     | 1   | 3    | 6.0   | 6.0  | 3.0 | NaN | 200.0 | 79.0 | -3.6 | 1021.8 | -19.1 | 0.0  | NNE | 1.0  | Dingling |
| 4 | 5  | 2013 | 3     | 1   | 4    | 5.0   | 5.0  | 3.0 | NaN | 200.0 | 81.0 | -3.5 | 1022.3 | -19.4 | 0.0  | N   | 2.1  | Dingling |

Figure A1

*Preview of the Beijing Multi-Site Air Quality Dataset (First Five Observations)*

[7] ✓ 0s df.shape

|              |
|--------------|
| (420768, 18) |
|--------------|

Figure A2

*Dataset Dimensions and Variable Information*

```
[8]
✓ Os df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 420768 entries, 0 to 420767
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  -
0    No           420768 non-null  int64
1    year         420768 non-null  int64
2    month        420768 non-null  int64
3    day          420768 non-null  int64
4    hour         420768 non-null  int64
5    PM2.5        412029 non-null  float64
6    PM10         414319 non-null  float64
7    SO2          411747 non-null  float64
8    NO2          408652 non-null  float64
9    CO           400067 non-null  float64
10   O3           407491 non-null  float64
11   TEMP         420370 non-null  float64
12   PRES         420375 non-null  float64
13   DEWP         420365 non-null  float64
14   RAIN         420378 non-null  float64
15   wd           418946 non-null  object
16   WSPM         420450 non-null  float64
17   station      420768 non-null  object
dtypes: float64(11), int64(5), object(2)
memory usage: 57.8+ MB
```

Figure A3

*Dataset Structure Showing Variable Names, Data Types, and Non-Null Observations*

[9] ✓ Os df.describe().T

|              | count    | mean         | std          | min       | 25%     | 50%     | 75%      | max     |
|--------------|----------|--------------|--------------|-----------|---------|---------|----------|---------|
| <b>No</b>    | 420768.0 | 17532.500000 | 10122.116943 | 1.0000    | 8766.75 | 17532.5 | 26298.25 | 35064.0 |
| <b>year</b>  | 420768.0 | 2014.662560  | 1.177198     | 2013.0000 | 2014.00 | 2015.0  | 2016.00  | 2017.0  |
| <b>month</b> | 420768.0 | 6.522930     | 3.448707     | 1.0000    | 4.00    | 7.0     | 10.00    | 12.0    |
| <b>day</b>   | 420768.0 | 15.729637    | 8.800102     | 1.0000    | 8.00    | 16.0    | 23.00    | 31.0    |
| <b>hour</b>  | 420768.0 | 11.500000    | 6.922195     | 0.0000    | 5.75    | 11.5    | 17.25    | 23.0    |
| <b>PM2.5</b> | 412029.0 | 79.793428    | 80.822391    | 2.0000    | 20.00   | 55.0    | 111.00   | 999.0   |
| <b>PM10</b>  | 414319.0 | 104.602618   | 91.772426    | 2.0000    | 36.00   | 82.0    | 145.00   | 999.0   |
| <b>SO2</b>   | 411747.0 | 15.830835    | 21.650603    | 0.2856    | 3.00    | 7.0     | 20.00    | 500.0   |
| <b>NO2</b>   | 408652.0 | 50.638586    | 35.127912    | 1.0265    | 23.00   | 43.0    | 71.00    | 290.0   |
| <b>CO</b>    | 400067.0 | 1230.766454  | 1160.182716  | 100.0000  | 500.00  | 900.0   | 1500.00  | 10000.0 |
| <b>O3</b>    | 407491.0 | 57.372271    | 56.661607    | 0.2142    | 11.00   | 45.0    | 82.00    | 1071.0  |
| <b>TEMP</b>  | 420370.0 | 13.538976    | 11.436139    | -19.9000  | 3.10    | 14.5    | 23.30    | 41.6    |
| <b>PRES</b>  | 420375.0 | 1010.746982  | 10.474055    | 982.4000  | 1002.30 | 1010.4  | 1019.00  | 1042.8  |
| <b>DEWP</b>  | 420365.0 | 2.490822     | 13.793847    | -43.4000  | -8.90   | 3.1     | 15.10    | 29.1    |
| <b>RAIN</b>  | 420378.0 | 0.064476     | 0.821004     | 0.0000    | 0.00    | 0.0     | 0.00     | 72.5    |
| <b>WSPM</b>  | 420450.0 | 1.729711     | 1.246386     | 0.0000    | 0.90    | 1.4     | 2.20     | 13.2    |

Figure A4

*Summary Statistics of the Numerical Variables*

[10] ✓ Os

df.isnull().sum()

|         | 0     |
|---------|-------|
| No      | 0     |
| year    | 0     |
| month   | 0     |
| day     | 0     |
| hour    | 0     |
| PM2.5   | 8739  |
| PM10    | 6449  |
| SO2     | 9021  |
| NO2     | 12116 |
| CO      | 20701 |
| O3      | 13277 |
| TEMP    | 398   |
| PRES    | 393   |
| DEWP    | 403   |
| RAIN    | 390   |
| wd      | 1822  |
| WSPM    | 318   |
| station | 0     |

dtype: int64

Figure A5

*Summary of Missing Values Across Dataset Variables*

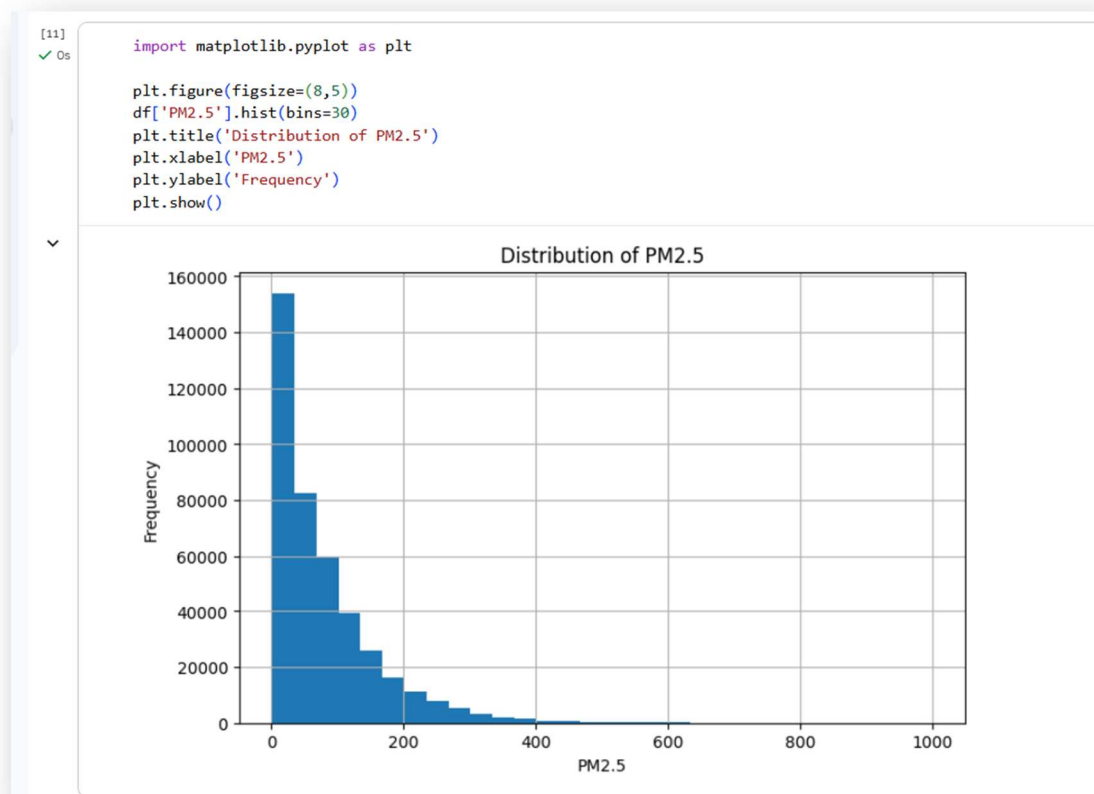


Figure A6

*Distribution of PM2.5 Concentration (Target Variable)*

## Appendix B

### GitHub Repository Structure

The GitHub repository was created to ensure reproducibility of the proposed study. It contains the project README, repository structure, dataset source information, and supporting files required for the capstone project.

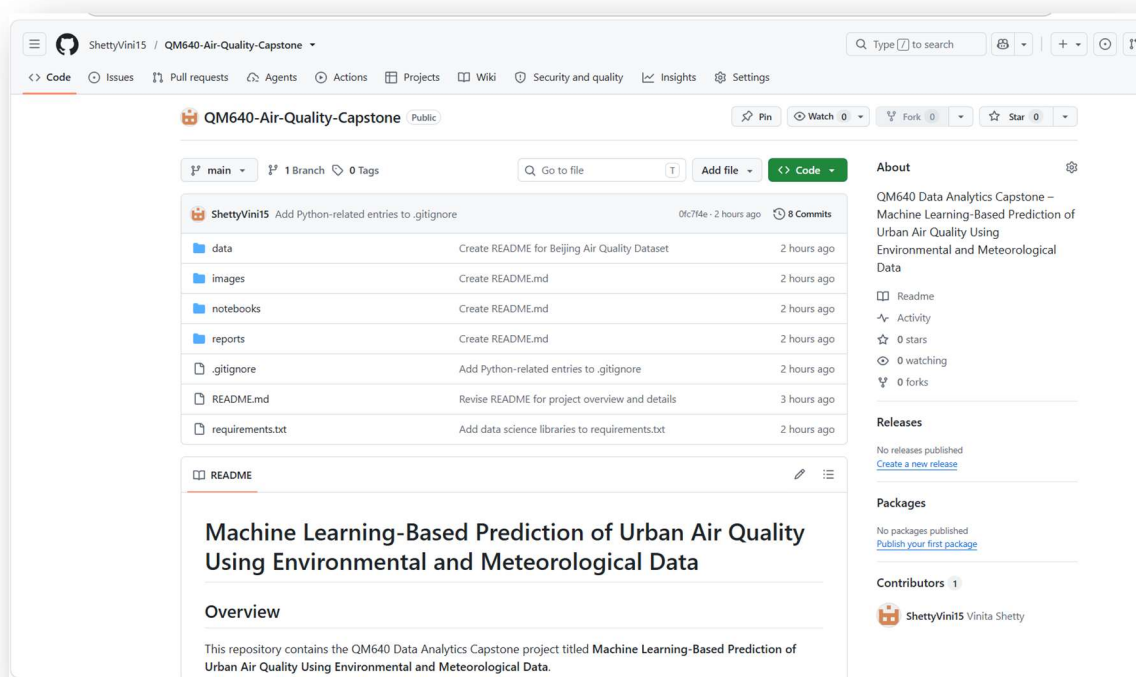


Figure B1

*Public GitHub Repository for the QM640 Air Quality Capstone Project*



## Project Repository Structure

```
QM640-Air-Quality-Capstone
|
├── data
|   └── README.md
|
├── notebooks
|   └── README.md
|
├── reports
|   └── README.md
|
├── images
|   └── README.md
|
├── README.md
├── requirements.txt
└── .gitignore
```

## Appendix C

### Research Hypotheses

**Table C1**

*Mapping of Research Questions to Research Hypotheses*

| Research Question | Null Hypothesis (H <sub>0</sub> )                                                                   | Alternative Hypothesis (H <sub>1</sub> )                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>RQ1</b>        | Environmental and meteorological variables do not significantly influence PM2.5 concentration.      | Environmental and meteorological variables significantly influence PM2.5 concentration.                           |
| <b>RQ2</b>        | Machine learning regression models do not accurately predict PM2.5 concentration.                   | Machine learning regression models accurately predict PM2.5 concentration.                                        |
| <b>RQ3</b>        | There is no significant difference in the predictive performance of the selected regression models. | At least one regression model demonstrates significantly better predictive performance than the others.           |
| <b>RQ4</b>        | Environmental and meteorological variables contribute equally to PM2.5 prediction.                  | Certain environmental and meteorological variables contribute more significantly to PM2.5 prediction than others. |

**Appendix D**  
**Proposed 10-Week Project Timeline**

| <b>Week</b>    | <b>Planned Activity</b>                                                                |
|----------------|----------------------------------------------------------------------------------------|
| <b>Week 1</b>  | Literature review, problem definition, dataset verification, GitHub repository setup   |
| <b>Week 2</b>  | Data preprocessing, cleaning, handling missing values, exploratory data analysis (EDA) |
| <b>Week 3</b>  | Feature selection, correlation analysis, and feature engineering                       |
| <b>Week 4</b>  | Development of Linear Regression and Decision Tree Regression models                   |
| <b>Week 5</b>  | Development of Random Forest, Gradient Boosting, and XGBoost models                    |
| <b>Week 6</b>  | Hyperparameter tuning, 5-fold cross-validation, and model refinement                   |
| <b>Week 7</b>  | Model evaluation using MAE, RMSE, and $R^2$ ; comparison of predictive performance     |
| <b>Week 8</b>  | Feature importance analysis, SHAP explainability, and interpretation of results        |
| <b>Week 9</b>  | Development of recommendations, documentation of findings, and report preparation      |
| <b>Week 10</b> | Final review, proofreading, GitHub updates, and capstone submission                    |